Lead Quality Optimization: Path to 20% Payout Bonus.

Analytics Case Study

Power BI

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Executive Summary

→ Objective
Increase lead quality from **8.1% → 9.6% (+20%)** to unlock **$33 CPL bonus** (current gap: **1.5%**).

→ Key Insights

1) Stable but insufficient trend: Lead quality remains relatively flat with slight recent improvement, but not enough to reach target organically.

2) Strong segment variation (key driver):

- **Partners:** *advertise.com* significantly outperforms others (e.g., Google, Yahoo).
- **Campaigns:** Debt-focused campaigns show higher lead quality vs generic segments.
- **Keywords:** High-intent search terms (debt-related) drive better conversions.
- **Ad Placement:** Top-right placements outperform top-left, indicating position impacts intent.

→ Opportunity to Reach 9.6%

1. Reallocate spend toward high-performing partners, campaigns, and keywords
2. Reduce low-quality traffic sources (underperforming partners & generic campaigns)
3. Optimize placements toward higher-performing positions
4. Reallocating spend toward top-performing segments can potentially close the 1.5% gap and achieve the 9.6% target.
5. Data quality signals (e.g., phone/address match scores) can further improve lead filtering and should be incorporated in future optimization.

Data Cleaning

Before conducting the analysis, the dataset was cleaned and structured to ensure accuracy and reliability of insights.

- **Resolved data inconsistencies:** Identified misclassified entries in the *Campaign* column (where state values were incorrectly recorded) and standardized them to **"content"** to maintain consistent categorization.
- **Standardized key dimensions:** Cleaned and harmonized categorical fields such as Partner, Marketing Campaign, Ad Placement, and Keywords to enable meaningful segmentation.
- **Reduced noise in data:** Removed irregularities and ensured uniform grouping across variables to prevent skewed analysis.
- **Feature readiness:** Prepared the dataset for analysis by ensuring all key variables were structured for accurate aggregation and comparison.

Feature engineering

To make the analysis more actionable and aligned with business objectives, key derived features were created:

- **is_good_eval:**
  A binary indicator to identify high-quality leads (e.g., closed or strong intent signals).
  Used as a proxy for success to evaluate campaign, keyword, and partner performance.
- **is_quality:**
  Represents the overall lead quality metric, serving as the primary KPI across the analysis for comparing performance across segments.
- **Time-based structuring (week_start):**
  Standardized date fields into weekly buckets to enable trend analysis and week-over-week performance tracking.

Data Processing Logic (Summary)

```python
# Overall quality (includes Unknown as 0)
df['is_quality'] = df['lead_category'].apply(
    lambda x: 1 if x in ['Closed', 'Good'] else 0
)

# Conditional quality (ONLY for evaluated leads)
df['is_good_eval'] = df['lead_category'].apply(
    lambda x: 1 if x in ['Closed', 'Good'] else (0 if x == 'Bad' else None)
)
```

- Creates key **lead quality indicators** (is_quality, is_good_eval) to classify good vs bad leads
- Builds **conversion metric (is_closed)** to track successful outcomes
- Standardizes fields like **debt level** for consistent grouping
- Converts **date fields** to enable time-based analysis

- Enables **accurate measurement** of lead quality and performance
- Provides a clear **success metric aligned with business goals**
- Ensures **clean, reliable segmentation** across campaigns and partners
- Establishes a **baseline (8.11%)** to benchmark improvements

```python
# Global Feature Engineering
# Create the success metric: 1 if Closed, 0 otherwise
df['is_closed'] = (df['callstatus'] == 'Closed').astype(int)

# Standardize debt labels for cleaner plotting
df['debtlevel'] = df['debtlevel'].replace('More_than_100000', '100000+')

# Ensure date is a datetime object
df['date'] = pd.to_datetime(df['date'])

# Calculate global baseline for reference lines
global_baseline = df['is_closed'].mean() * 100

print(f"Dataset Loaded: {len(df)} leads")
print(f"Global Baseline Quality: {global_baseline:.2f}%")
```

```
Dataset Loaded: 3021 leads
Global Baseline Quality: 8.11%
```

Data Processing Logic (Summary)

- Identifies incorrect values in the marketing campaign column where entries contain **“state”** .
- Replaces these values with **“content”** to ensure consistent campaign categorization.
- Classifies leads into good vs others based on their status.
- Separates evaluated leads (good vs bad) and excludes unknown cases for accurate analysis.
- Filters the dataset to keep only valid, evaluated leads.
- Calculates average lead quality by partner to compare performance across traffic sources.

```
df.loc[
    df['marketingcampaign'].str.lower().str.contains('state', na=False),
    'marketingcampaign'
] = 'content'
```

```
# Overall quality (includes Unknown as 0)
df['is_quality'] = df['lead_category'].apply(
    lambda x: 1 if x in ['Closed', 'Good'] else 0
)

# Conditional quality (ONLY for evaluated leads)
df['is_good_eval'] = df['lead_category'].apply(
    lambda x: 1 if x in ['Closed', 'Good'] else (0 if x == 'Bad' else None)
)
```

```
df_eval = df[df['is_good_eval'].notna()]
```

```
df.groupby('partner')['is_quality'].mean()
```

✓ 0.0s

Key Data Challenges & Fixes

- **Inconsistent Campaign Labeling**
  Campaign column contained incorrect values (e.g., *state* entries mixed with campaign names), leading to inaccurate segmentation
  Fix: Standardized values by replacing such entries with **"content"**, ensuring consistent campaign grouping

- **No Clear Definition of "Good Lead"**
  Dataset did not explicitly identify high-quality leads
  Fix: Created a binary indicator **is_good_eval** to define and track good leads as a proxy for conversions

- **Unstructured Date Field for Trend Analysis**
  Time-based analysis was not directly possible due to improper formatting
  Fix: Converted **week_start** into a proper datetime format to enable week-over-week analysis

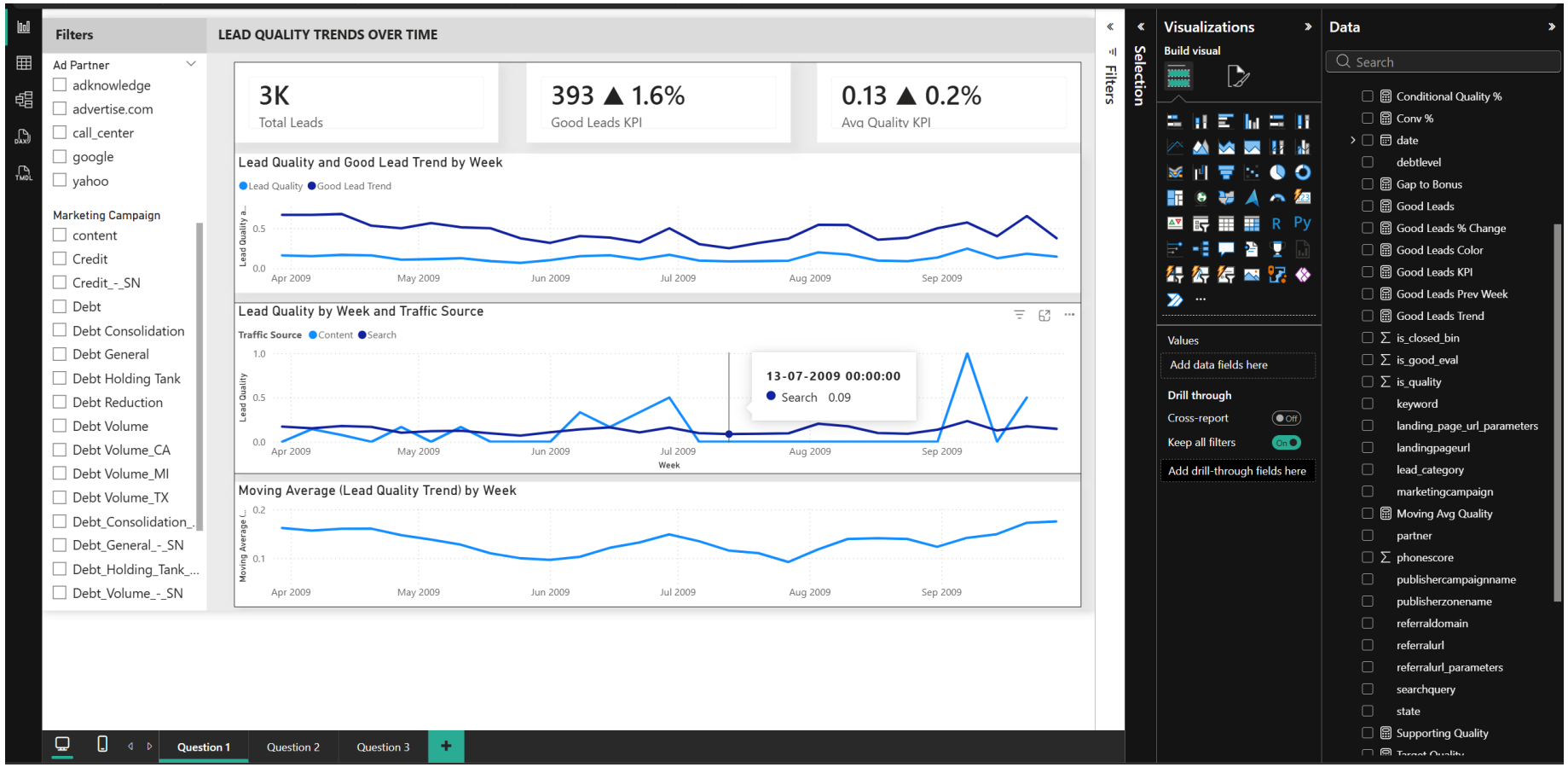
Analytical Challenges & Approach

- **High Variability Across Segments**
  Performance varied significantly across campaigns, partners, and keywords, making it difficult to identify clear drivers.
  Approach: Conducted multi-dimensional analysis to isolate high-performing vs low-performing segments and uncover actionable patterns.

- **Balancing Lead Quality vs Volume**
  Efforts to improve lead quality risked reducing overall lead volume.
  Approach: Focused on traffic mix optimization—scaling high-quality segments while minimizing low-quality, high-volume sources.

- **Identifying True Performance Drivers**
  Raw data did not clearly reveal the factors influencing lead quality.
  Approach: Developed derived metrics such as is_quality and is_good_eval to enable structured evaluation and data-driven decision-making.

Q.1 Are we seeing any lead quality trends over time (improving, declining)? Are they statistically significant?



- Lead quality remains relatively stable with minor fluctuations over time
- Slight improvement observed in later weeks, but no strong upward trend
- Variations appear inconsistent rather than systematic → No inherent improvement over time—performance is driven by segment mix, not time-based optimization.

Statistical Validation of Lead Quality Trends

```
from scipy.stats import ttest_ind

t_stat, p_value = ttest_ind(early, late)

print("T-stat:", t_stat)
print("P-value:", p_value)
```

```
T-stat: 0.04703243017342879
P-value: 0.962490491404352
```

```
print("Early quality:", early.mean())
print("Late quality:", late.mean())
```

```
Early quality: 0.13037723362011913
Late quality: 0.12980132450331125
```

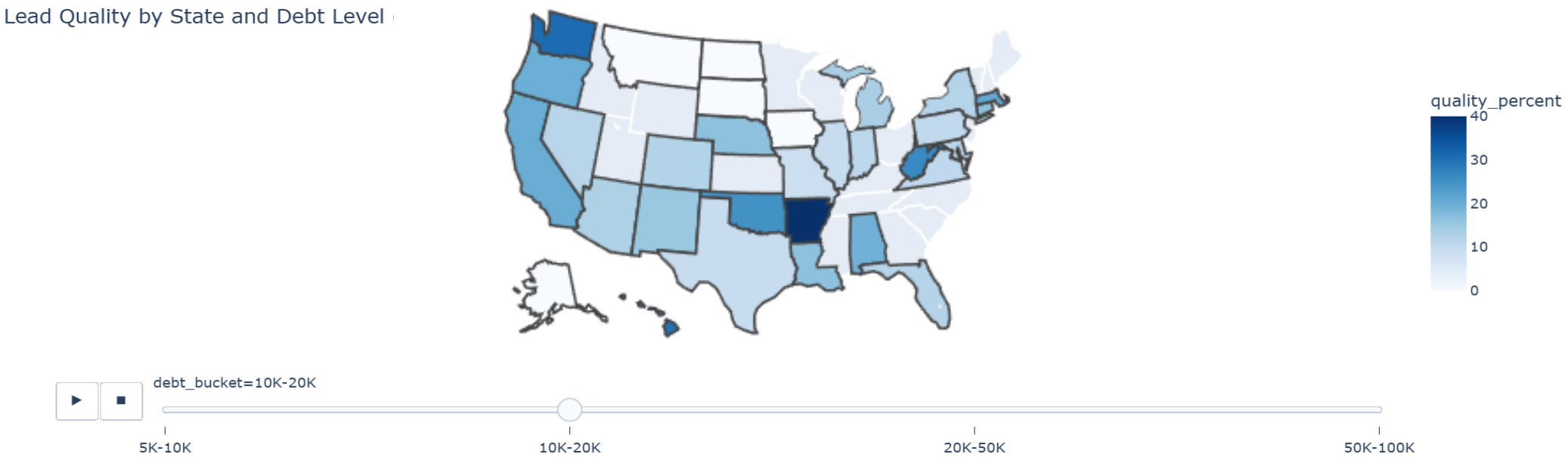
- Conducted statistical test (t-test) between early vs late periods
- High p-value (~0.96) indicates no statistically significant difference in lead quality over time

Q. 2 What can we learn about the drivers of "lead quality" from this dataset? What segments: 1) where the ad was shown, 2) what kind of person filled out the ad 3) what kind of ad did they see have differing lead quality rates?

- **Campaigns**: Debt-focused campaigns outperform generic ones
- **Partners:** Significant variation—some sources deliver consistently higher-quality leads
- **Ad Placement**: Certain placements (e.g., top-right) perform better
- **User Intent (Debt Level):** Higher debt segments show better conversion quality

Lead quality is driven by who we target, where ads appear, and traffic source quality

Map Visualization for Debt Level



Outcome: Certain states show higher lead quality, indicating geographic targeting potential

Q.3 If the advertiser says they will increase our CPL by 20% (i.e., $30 to $33) if we increase our lead quality by 20% (i.e., from 8.0% to 9.6%), do we see any opportunities to do that here? What kinds of things could we do?

Based on the analysis, there are clear opportunities to increase lead quality from ~8% to the target 9.6% by optimizing key drivers such as campaigns, partners, keywords, and ad placements. The dashboard highlights significant variation in performance across these dimensions, which we can leverage to improve overall quality.

- From a **campaign perspective**, we see that certain campaigns like *Debt Volume* and *Debt* deliver significantly higher lead quality compared to others. This suggests that reallocating budget toward these high-performing campaigns, while reducing investment in low-quality campaigns, can directly improve overall performance.

- At the **partner level**, *advertise.com* stands out as the strongest performer, whereas partners like *google* and *yahoo* lag behind in quality. This indicates an opportunity to shift more volume toward high-quality partners and optimize or limit exposure to underperforming ones.

- Looking at **ad placements**, we observe that *Top Right (300x250)* placements outperform others, suggesting that prioritizing better-performing placements can further enhance lead quality.

- Finally, from a **keyword standpoint**, a small set of keywords is driving a disproportionately high number of good leads. Focusing spend on these high-performing keywords and eliminating inefficient ones can significantly improve conversion efficiency. The target can be achieved by improving the composition of leads rather than increasing volume.

Real-World Relevance (FinTech)

In fintech, growth isn't driven by more leads—it's driven by better leads that actually convert:

LendingTree improved performance by prioritizing high-intent users
→ Resulted in 2–3x higher conversion rates and more efficient spend

Credit Karma used data to match users with relevant financial products
→ Drove 20–30% improvement in engagement and approvals

SoFi focused on acquiring financially qualified users rather than scaling volume
→ Reduced acquisition costs while improving ROI

This project applies the same principle:
→ Not all traffic is equal—performance improves by shifting toward high-intent segments

A small lift in quality (~1–2%) can meaningfully impact revenue—reflected here through a 20% increase in CPL

THANK YOU